COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Tutorial 2 — Signed, Unsigned and Fast Multiplication
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Section A: Unsigned Multiplication

• 1a) 1101 × 1011: partial products 1101 + 0000 + 110100 + 1101000 → 10001111. Answer: 10001111 (13 × 11 = 143)
• 1b) 10101 × 0110: partial products 000000 + 101010 + 1010100 + 0000000 → 1111110. Answer: 1111110 (21 × 6 = 126)
• 2a) 12 × 5: 12 = 1100, 5 = 0101; binary product 1100 × 0101 = 111100; back to decimal: 32 + 16 + 8 + 4 = 60. Answer: 60
• 2b) 23 × 9: 23 = 10111, 9 = 1001; binary product = 11001111; decimal: 128 + 64 + 8 + 4 + 2 + 1 = 207. Answer: 207
• 3) 1110 (M) × 1001 (Q), 4-bit registers, add-and-shift hardware algorithm:
• Init: A = 0000, MQ = 1001, MD = 1110, counter = 4
• Step 0: MQ0 = 1 → A = A + MD = 0000 + 1110 = 1110 (product 1110 1001); logical shift right → 0111 0100
• Step 1: MQ0 = 0 → no add; shift → 0011 1010
• Step 2: MQ0 = 0 → no add; shift → 0001 1101
• Step 3: MQ0 = 1 → A = A + MD = 0001 + 1110 = 1111 (1111 1101); final shift → 0111 1110
• Product = 01111110 = 126 (check: 14 × 9 = 126 ✓)

Section B: Signed Multiplication (Booth's Algorithm)

• a) (+7) × (+3): M = 0111, -M = 1001, Q = 0011, A = 0000, Q-1 = 0
• Cycle 1: Q0Q-1 = 10 → A = A - M = 1001; shift → A = 1100, Q = 1001, Q-1 = 1
• Cycle 2: Q0Q-1 = 11 → no op; shift → A = 1110, Q = 0100, Q-1 = 1
• Cycle 3: Q0Q-1 = 01 → A = A + M = 0101; shift → A = 0010, Q = 1010, Q-1 = 0
• Cycle 4: Q0Q-1 = 00 → no op; shift → A = 0001, Q = 0101, Q-1 = 0
• A + Q = 0001 0101 = 00010101 = 21 (7 × 3 = 21 ✓)
• b) (-5) × (+4): M = 1011, -M = 0101, Q = 0100, A = 0000, Q-1 = 0
• Cycle 1: Q0Q-1 = 00 → shift → A = 0000, Q = 0010, Q-1 = 0
• Cycle 2: Q0Q-1 = 00 → shift → A = 0000, Q = 0001, Q-1 = 0
• Cycle 3: Q0Q-1 = 10 → A = A - M = 0000 + 0101 = 0101; shift → A = 0010, Q = 1000, Q-1 = 1
• Cycle 4: Q0Q-1 = 01 → A = A + M = 0010 + 1011 = 1101; shift → A = 1110, Q = 1100, Q-1 = 0
• A + Q = 1110 1100 = -20 in 2's complement (-5 × 4 = -20 ✓)
• c) (-7) × (-3) with 5-bit registers: M = 10001, -M = 01111, Q = 11101, A = 00000, Q-1 = 0
• Cycle 1: 10 → A = A - M = 01111; shift → A = 00111, Q = 11110, Q-1 = 1
• Cycle 2: 01 → A = A + M = 10110; shift → A = 11011, Q = 01111, Q-1 = 0
• Cycle 3: 10 → A = A - M = 01010; shift → A = 00101, Q = 00111, Q-1 = 1
• Cycle 4: 11 → no op; shift → A = 00010, Q = 10011, Q-1 = 1
• Cycle 5: 11 → no op; shift → A = 00001, Q = 01001, Q-1 = 1
• Product (A, Q) = 00001 01001 = 000010101 = +21 (-7 × -3 = +21 ✓)
• d) M = 110011 (-13), Q = 101110 (-18): A = 000000, -M = 001101, Q-1 = 0
• Cycle 1: 00 → no op; shift → A = 000000, Q = 010111, Q-1 = 0
• Cycle 2: 10 → A = A - M = 001101; shift → A = 000110, Q = 101011, Q-1 = 1
• Cycle 3: 11 → no op; shift → A = 000011, Q = 010101, Q-1 = 1
• Cycle 4: 11 → no op; shift → A = 000001, Q = 101010, Q-1 = 1
• Cycle 5: 01 → A = A + M = 110100; shift → A = 111010, Q = 010101, Q-1 = 0
• Cycle 6: 10 → A = A - M = 000111; shift → A = 000011, Q = 101010, Q-1 = 1

• Product = 000011 101010 = 000011101010 = +234 (-13 × -18 = +234 ✓)

Section C: Fast Multiplication (Karatsuba)

• a) 1011 × 1101 (4-bit): split X = 10|11, Y = 11|01 (halves of 2 bits)

• z2 = Xh·Yh = 10 × 11 = 110 (6); z0 = Xl·Yl = 11 × 01 = 11 (3)

• z1 = (Xh + Xl)(Yh + Yl) - z2 - z0 = (101)(100) - 110 - 11 = 10100 - 1001 = 1011 (11)

• Product = z2·2^4 + z1·2^2 + z0 = 1100000 + 101100 + 11 = 10001111 = 143 (11 × 13 = 143 ✓)

• b) 54 × 87: 54 = 00110110, 87 = 01010111 (8-bit); split into 4-bit halves: Xh = 0011 (3), Xl = 0110 (6), Yh = 0101 (5), Yl = 0111 (7)

• z2 = 3 × 5 = 15; z0 = 6 × 7 = 42; z1 = (3 + 6)(5 + 7) - 15 - 42 = 9 × 12 - 57 = 108 - 57 = 51

• Product = 15 × 256 + 51 × 16 + 42 = 3840 + 816 + 42 = 4698 (54 × 87 = 4698 ✓)

• In binary: 4698 = 1001001011010

• c) X = 11011011, Y = 10111001 setup: Xh = 1101, Xl = 1011, Yh = 1011, Yl = 1001

• Three recursive terms: z2 = Xh·Yh = 1101 × 1011; z0 = Xl·Yl = 1011 × 1001; z1 = (Xh + Xl)(Yh + Yl) - z2 - z0 = (11000)(10100) - z2 - z0

• Final combination: Product = z2·2^8 + z1·2^4 + z0